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From Quantum Computing Algorithms to Human Health Applications: Tunnelling through the Disciplinary Barriers

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Abstract

We describe a collaborative research project spanning the disciplines of quantum hardware, quantum algorithms, conventional computational chemistry, synthetic medicinal chemistry and life sciences. Our project seeks to demonstrate an impact of quantum computing on human health. It is one of several funded by Wellcome Leap as part of their Quantum for Bio Program. We aim to use quantum computing to guide the discovery of covalent inhibitors for the treatment of the rare disease myotonic dystrophy type I. Modelling the formation of a chemical bond in the context of the rest of the inhibitor and the protein is a challenge, because of the size of the systems of interest, the intricacies of the electronic structure where the importance of electron correlation may vary between reactant and transition states, and the extensive sampling of configurational space that is often required. We are developing an efficient hybrid quantum-classical framework to address the accuracy and scalability limitations of conventional computational drug design approaches. A quantum-enhanced density functional approximation is developed based on data from computation on neutral-atom quantum hardware, which can then be deployed within conventional quantum mechanics/molecular mechanics calculations. Our quantum-enhanced framework can be systematically updated as more capable quantum processors become available, heralding the prospect of using quantum-derived data to train improved density functional approximations and, ultimately, to support a wide range of drug design efforts.

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Keywords

drug discovery, covalent inhibitor, quantum-enhanced density functional theory, QM/MM

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Abstract: We describe a collaborative research project spanning the disciplines of quantum hardware, quantum algorithms, conventional computational chemistry, synthetic medicinal chemistry and life sciences. Our project seeks to demonstrate an impact of quantum computing on human health. It is one of several funded by Wellcome Leap as part of their Quantum for Bio Program. We aim to use quantum computing to guide the discovery of covalent inhibitors for the treatment of the rare disease myotonic dystrophy type I. Modelling the formation of a chemical bond in the context of the rest of the inhibitor and the protein is a challenge, because of the size of the systems of interest, the intricacies of the electronic structure where the importance of electron correlation may vary between reactant and transition states, and the extensive sampling of configurational space that is often required. We are developing an efficient hybrid quantum-classical framework to address the accuracy and scalability limitations of conventional computational drug design approaches. A quantum-enhanced density functional approximation is developed based on data from computation on neutral-atom quantum hardware, which can then be deployed within conventional quantum mechanics/molecular mechanics calculations. Our quantum-enhanced framework can be systematically updated as more capable quantum processors become available, heralding the prospect of using quantum-derived data to train improved density functional approximations and, ultimately, to support a wide range of drug design efforts.

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Introduction

"Seemingly impossible results on seemingly impossible timelines" is the stated aim of Wellcome Leap [1], and this certainly seems apposite in the fast-moving field of quantum computing. Wellcome Leap's overarching aim is evident in its US$40M + US$10M Supported Challenge Program in Quantum for Bio (Q4Bio), where the remit is to demonstrate the potential of quantum computing in solving human health challenges. We are one of only six teams worldwide in the third phase of the Q4Bio Challenge Program. In this paper, prompted in part by the final session at a Royal Society discussion meeting in early October 2025 on "*Quantum computing in materials and molecular sciences*" [2], which explored some of the collaborative challenges in the field, we discuss our Q4Bio project and reflect on how it has accelerated inter-disciplinary collaboration to connect quantum computing more closely to real-world applications.

Overview of the Healthcare Challenge

We aim to accelerate the drug discovery process for covalent inhibitors, with the rare disease myotonic dystrophy [3] as an exemplar. Drug discovery is a notoriously long, costly, and high-risk process, with a significant rate of failure [4]. These challenges are amplified in the context of rare diseases, where limited market size often discourages private sector investment.

Myotonic dystrophy type I (DM1), is a severe, progressive, debilitating condition which affects multiple systems including skeletal muscle, heart and brain. It is the most common adult form of muscular dystrophy, with an estimated 6,500 people affected in the UK. There is currently no treatment available. The presence of an expanded CTG trinucleotide repeat in the 3'-untranslated region of the *DMPK* gene is characteristic of DM1 [5]. This gene expansion results in long CUG triplet repeats containing mRNA molecules that form discrete structures (foci) in cell nuclei that can be visualised via *in situ* hybridisation probes. The CUG-expanded mRNA is retained in the nucleus and hence is not translated into protein. No 'druggable' protein target had been identified when the team from the University of Nottingham started our DM1 drug discovery programme, and 15 years ago we developed a medium-throughput phenotypic screening assay that allowed us to screen drug and druglike small molecule libraries to find hit molecules that were capable of reducing nuclear foci numbers in patient-derived cell lines. In 2014, we showed that kinase inhibitors were worthy of further investigation [6]. Subsequent efforts, in collaboration with GlaxoSmithKline plc (who provided access to their Published Kinase Inhibitor Set) and with Cellzome (who provided access to their Kinobead kinase screening assay), allowed us to identify cyclin-dependent kinase 12 (CDK12) as a novel, druggable protein target for DM1 [7].

Building on this discovery, we have secured two major drug discovery grants (Wellcome Trust's Seeding Drug Discovery and MRC DPFS) to develop inhibitors of CDK12 that have high selectivity over other CDK family members, particularly CDK9, and other kinases in general. During our studies it became clear that a covalent inhibition strategy was promising as it provides a clear mechanism to achieve the desired levels of selectivity over CDK9 by exploiting the presence of a nucleophilic cysteine residue (Cys1039) in CDK12 that is absent in CDK9 [8]. However, creating such selective compounds remains time-consuming and expensive (standing at >15 years and >£6M), as the drug discovery process still requires the

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iterative synthesis and assay cycles. Current computational chemistry tools lack the precision needed to predict how subtle structural changes will influence drug potency and selectivity. As a result, hundreds of compounds must be synthesised and tested over several years to optimise selectivity and potency, with design decisions often based on limited predictive accuracy.

Through the Wellcome Leap project, our collaborative team is developing advanced methods to model covalent inhibition of CDK12 more accurately. These innovations are expected to accelerate the drug development process significantly. Importantly, this approach is not limited to myotonic dystrophy; it has potential applications in a broad range of conditions where CDK12 plays a role, including cancers [9] and other genetic neuromuscular disorders. Covalent inhibition is a powerful technique to improve potency and selectivity of drugs, and 30% of currently marketed drugs are covalent inhibitors [10]. These methods could transform covalent inhibitor design across drug discovery programs, offering a powerful strategy for tackling many hard-to-treat diseases.

The Classical Computational Challenge

The application of computational chemistry to biological problems for Healthcare has been propelled forward in the last quarter of century not only by the exponential growth of classical computing power (Moore's Law), but also by the development of algorithms that can leverage the immense computing power now available to humanity. The transformational power of this digital revolution has been recognised by Nobel Prizes for Chemistry in 1998 going to Kohn and Pople for work on computational chemistry, including density functional theory (DFT), and again in 2013 when the Nobel Prize was awarded to Karplus, Levitt and Warshel for their pioneering work on hybrid quantum mechanics/molecular mechanics (QM/MM) approaches for simulation of biological chemistry.

The computer revolution transformed our understanding of QM by using high-performance computing (HPC) facilities to simulate molecular systems. This has enabled transformational progress in drug discovery along with the molecular and material sciences. A workhorse of this revolution has been DFT, which at its heart relies on the approximation of the, as yet unknown, exchange-correlation functional, limiting its general applicability, in particular for cases where there are strong quantum effects. Thus, the routine use of DFT for guiding molecular discovery is stymied, as exact methods are only tractable, and therefore can only inform density-functional approximation (DFA) design, for relatively small systems. Algorithmic advances in quantum-enhanced DFT (QE-DFT) aims to usher in an analogous shift for quantum computing, making complex problems practically and routinely accessible, and meanwhile transitioning DFT into a new era of DFA design informed by simulations performed on a quantum computer. With this we aim to overcome well-known limitations of standard approaches for the design of covalent inhibitors.

Despite significant advances through application of classical approaches, covalent drug design presents a challenging frontier for classical computational resources, because only the most advanced and poorly scaling methods can capture the key interactions at the binding sites [11-14]. For example, QM/MM simulations, where we break down a large intractable problem into a classically treated region and a quantum mechanical region, the size of the quantum region needed to study covalent inhibitor ligand-protein binding (Figure

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1) severely limits the applicability of these methods, and computational chemists must make concessions through approximations which suffer spurious errors and numerical instabilities. This bottleneck in classical computations severely limits progress in the burgeoning field of covalent drug design, which has central relevance to the treatment of DM1, and any hard-to-treat disease which requires high potency, selectivity and long duration of action.

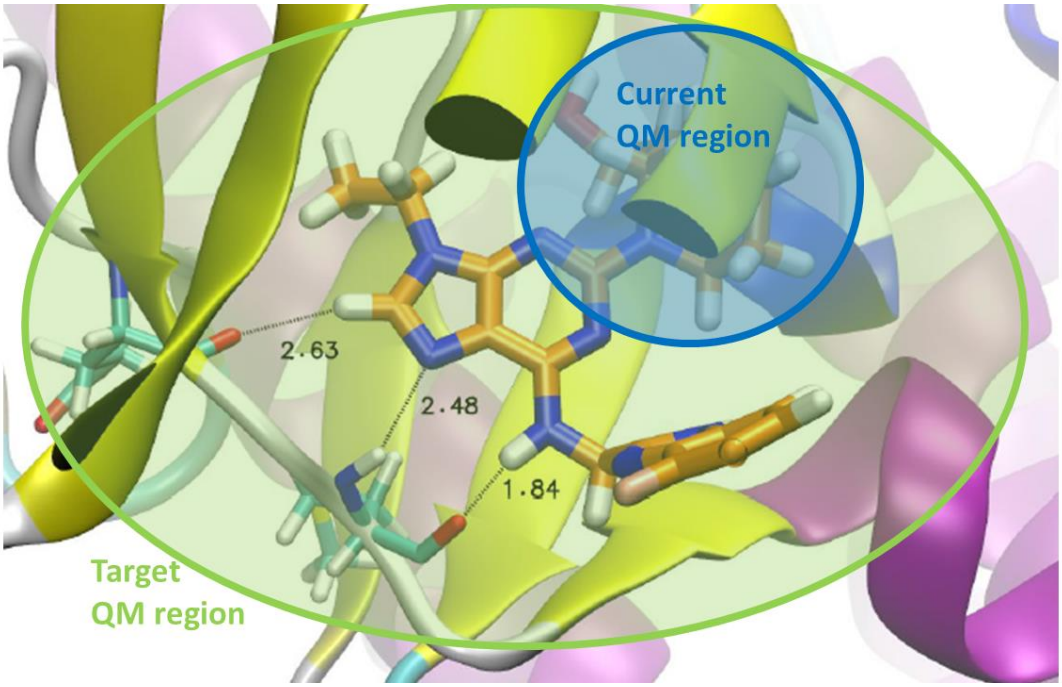

Figure 1. Hybrid QM/MM simulations involve the definition of a region treated quantum mechanically with the remainder of the system treated with an empirical molecular mechanics (MM) force field. The cost and scaling of current electronic structure methods means that the QM region is truncated quite severely (current QM region in blue *versus* target QM region in green). Ideally, the QM region would include the main chain of the CDK12 protein around the Cys1039 residue, the ligand itself and amino acids of the "hinge" region of the protein. Shown here are the Glu814 and Met816 residues from PDB structure 6B3E [15] binding a ligand from AstraZeneca plc.

Now, we stand on the brink of a new paradigm in computing. The use of quantum mechanical phenomena, entanglement and superposition, opens up a new more powerful way to simulate quantum mechanical problems, as are found in quantum chemistry for drug discovery. Quantum computing will bring about a step change in the way we simulate, understand and ultimately predict phenomena found in nature - that up until this point have remained elusive.

The Nottingham–Phasecraft–QuEra Framework

Undoubtedly, finding practical quantum advantage for a human health application of significance is an ambitious endeavour [16]. Our project is tackling both the accuracy and scalability limitations of computational drug design by development of an efficient hybrid quantum-classical framework (Figure 2). In our approach, improved accuracy is built-in from

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true quantum treatment of covalent interactions inherent to quantum computations, and scalability is ensured by optimally splitting the computational load between classical and quantum resources.

We are working on a novel pathway to quantum-accelerated computations of covalent inhibitors on near-term quantum devices by extending Phasecraft's advances in quantum algorithm design of low-depth and compact quantum circuits to molecular systems, informed by the specific hardware capabilities and features of QuEra's neutral-atom quantum computers. The existing pipeline comprises calculations using the following approaches: variational quantum eigensolver (VQE) [17], quantum-enhanced density functional theory (QE-DFT) [18, 19] and the well-established (on conventional computing hardware) technique of hybrid QM/MM simulations. VQE is a method that can be used to obtain stable energies of molecules on near-term quantum hardware and has been used during the pipeline development in this program to provide training data for QE-DFT. However, our framework is agnostic with respect to the underlying quantum energy solver used to extract the observables relevant to the QE-DFT component. As a result, it can be readily adapted to more scalable and resource-intensive techniques, such as quantum phase estimation, as they become viable in the future.

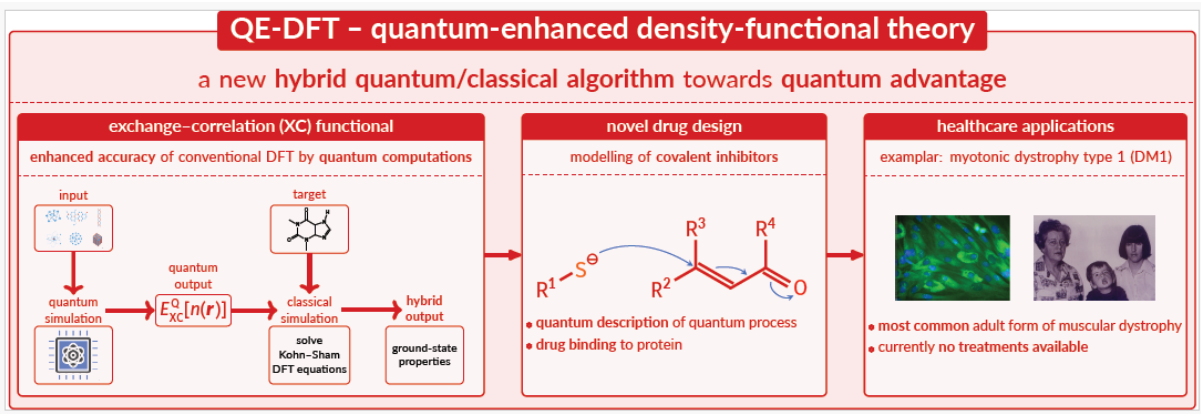

Figure 2. Overarching vision: the application of quantum computing to an important biological problem.

Limitations of current quantum hardware confine us to simulation of small molecular systems. One potential strategy for simulation of a large inhibitor molecule is a divide-and-conquer approach, such as density matrix embedding theory (DMET) [20]. However, the smallest fragmentation may be inadequate to capture the total energy of the system. Larger fragments do tend toward improved accuracy, but this comes at a size that is beyond switching in a quantum solver itself. Therefore, we need a different approach to scale to larger systems. With QE-DFT, we have been improving the DFAs that we fit, through fundamental and numerical enhancements to the routines. Furthermore, as we scale our VQE simulations, we obtain data that enables the training of improved DFAs. Importantly, in some cases, DFAs emerge that outperform their training data and come closer to the ground truth (where it is possible to calculate this exactly).

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In summary, our solution is embodied in the Nottingham–Phasecraft–QuEra (NPQ) framework for end-to-end hybrid quantum-classical simulation (Figure 3), and we are applying this to covalent drug discovery. The NPQ framework can take any molecular system and provide optimised circuits and quantum hardware resource estimates with orders of magnitude improvement against the state-of-the-art. The NPQ approach maximises flexibility, in order to be practical within diverse drug discovery pipelines and sustainable with advancing algorithmic and device capabilities. We can thus adapt our algorithmic choices to the chemical problem and available quantum resources. Such choices include the guess for the mathematical form of the solution (*ansatz*), filtering threshold for discarding insignificant interaction terms, and the mathematical functions used to discretize the electronic structure of interest (basis set), as well as the quantum training data input to QE-DFT.

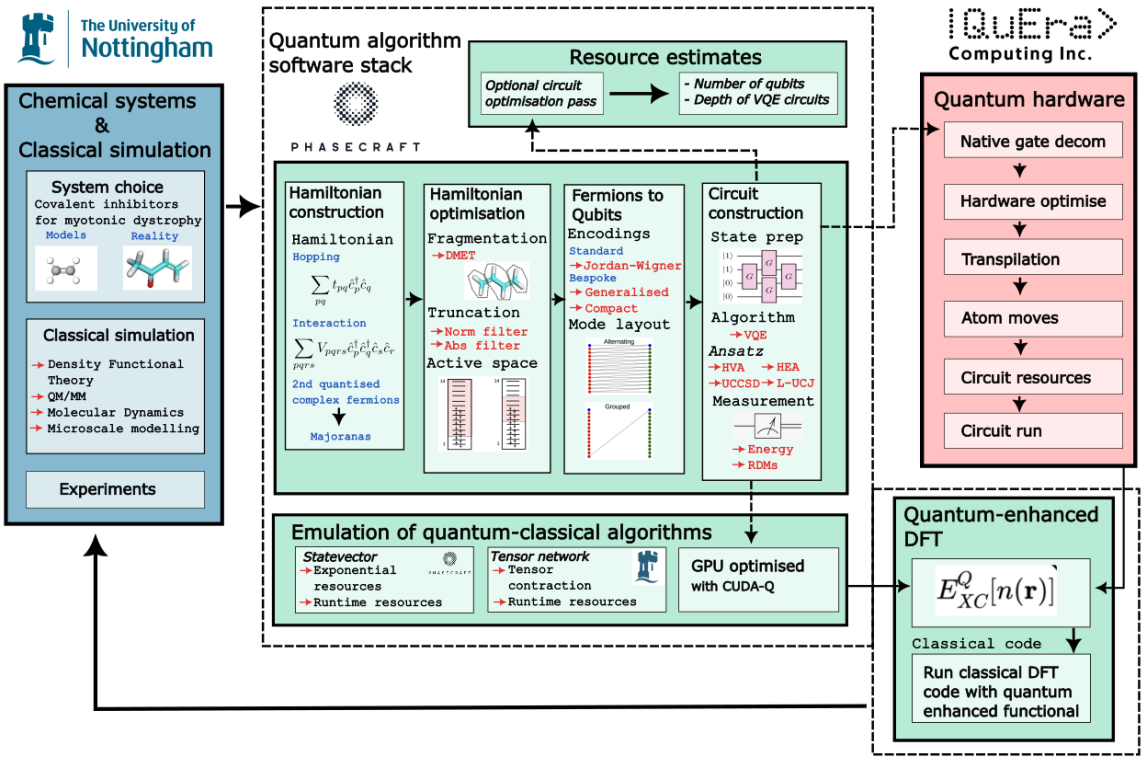

Figure 3. The Nottingham–Phasecraft–QuEra (NPQ) framework for hybrid quantum-classical computing. Nottingham brings expertise in the medicinal use case and classical biochemistry simulation interfacing with Phasecraft's quantum algorithm stack for efficient circuit preparation and QE-DFT. This in turn interfaces with QuEra's world-leading neutral-atom hardware. The loop is closed with QE-DFT then feeding back into the classical workflow and application to the DM1 drug discovery pipeline.

Interdisciplinary Teamwork

The Q4Bio challenge has been a highly interdisciplinary project, bringing scientists together from quantum algorithms and hardware, conventional quantum chemistry and synthetic medicinal chemistry. The Venn diagram in Figure 4 illustrates how the disciplines are intersecting. This represents a multi-person bridge from a human health application to

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deployment on hardware, where technically disparate expertise was drawn upon. Interestingly, materials physics became a useful leverage point, as quantum computing has advanced further in this field than biomedicine, such that techniques could be built on this foundation whilst providing a degree of overlap between knowledge bases.

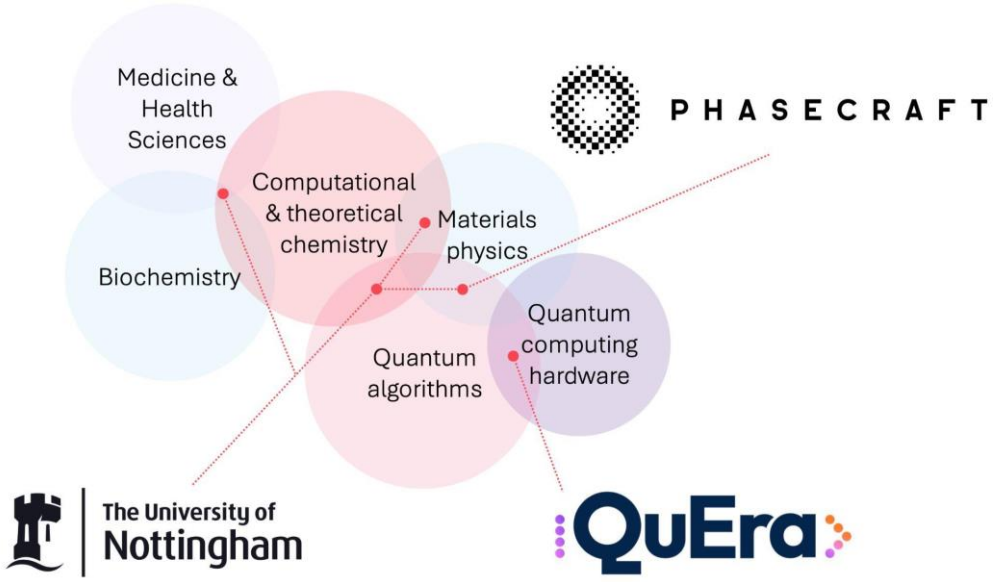

Figure 4. Venn diagram highlighting the interdisciplinary nature of the project, ranging from medicine and health sciences to realisation on quantum hardware, and the interfaces of technical disciplines where scientists from the partner organisations are working.

Our team brings together expertise from human health sciences and computational chemistry at the University of Nottingham (UoN), quantum algorithm development at Phasecraft Ltd. and quantum hardware at QuEra Computing Inc. UoN has a vibrant computational and theoretical chemistry section with PIs working in many areas of quantum chemistry and molecular simulation. There is also an advanced research programme to discover new drugs to treat the disease DM1, building on the discovery, several decades ago, of the genetic basis of the disease [5], with new compounds being synthesised by researchers in the School of Chemistry [8]. Phasecraft's main goal is to exploit near-term quantum computers to deliver quantum advantage. Proprietary technology and methods [18, 19] from within the company comprise some of the building blocks of our Q4Bio challenge. QuEra is the leading provider of quantum computers based on neutral atoms [21, 22].

We selected neutral atoms as the primary hardware modality due to their inherent scalability and the architectural flexibility they provide. QuEra's platform supports dynamic reconfiguration of qubit layouts through atom shuttling, enabling the architecture to be tailored to the specific characteristics of a given problem. Projected advances over the next five years, in particular, increases in qubit counts and improvements in gate fidelities, as well as incorporation of error correction into computational workflows are expected to expand the regime in which quantum processors can deliver practically meaningful insights. These developments in combination with the framework developed during the Q4Bio challenge program open up a pathway for neutral-atom quantum systems to contribute to problems of interest in drug-design pipelines in the long run.

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Technical Discussion

Michael acceptors are a class of chemical whose reactive chemical structure is ideally suited to covalent bond formation with amino-acid residues (specifically cysteine in our case). More chemically speaking, the addition of thiols to Michael acceptors (α,β-unsaturated carbonyl compounds and their derivatives) are key bond-forming processes in the burgeoning field of covalent drug design. These are challenging chemical reactions for quantum theory, and methods that accurately treat electron delocalisation and non-bonded interactions are essential. The inhibitors that we are investigating typically comprise over 30 heavy (*i.e.*, non-hydrogen) atoms and over 200 electrons. Subtle changes in chemical structure can lead to large differences in selectivity. For example, there is a pair of Michael acceptor inhibitors (Figure 5) where changing the nitrogen heterocycle in the main warhead from an azetidinyl, $–N(CH_2)_3$, group to a pyrrolidinyl, $–N(CH_2)_4$, group changes the selectivity between CDK9 and CDK12 from three-fold to zero.

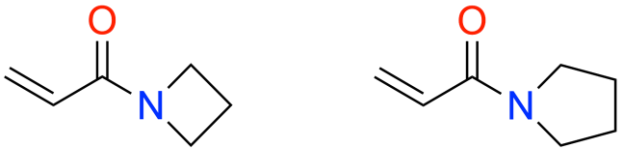

Figure 5. Chemical structures of two Michael acceptors: azetidinyl vinyl ketone (left) and pyrrolidinyl vinyl ketone (right).

Small chemical changes can also affect binding affinity in unanticipated ways. The α-fluoro-derivative in Figure 6 was anticipated to show improved potency, due to the electron-withdrawing nature of fluorine. However, more subtle electronic factors are clearly at play and capturing these would require simulations to be conducted both at scale and accuracy that is beyond the boundaries of what is achievable on a classical device.

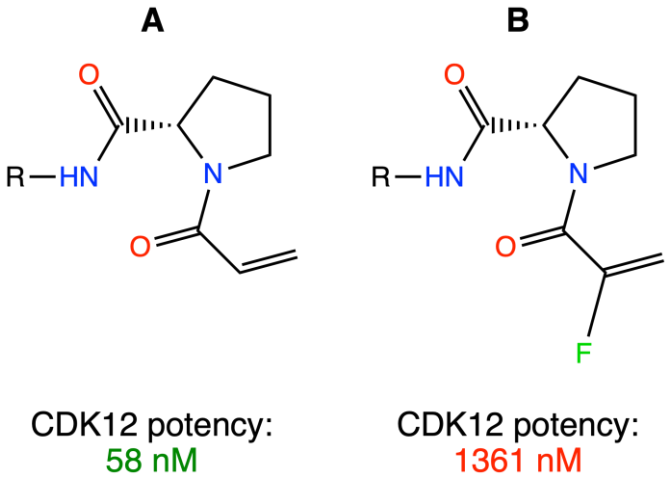

Figure 6. Modifying the hydrogen atom (compound **A**) bonded to the α-carbon atom to a fluorine atom (compound **B**) unexpectedly reduces potency by 20-fold from 58 nM to 1361 nM. Proprietary parts of the structures are depicted as R.

In general, Michael acceptors and the full inhibitor molecules are beyond direct emulation or running 'as is' on existing quantum devices. Therefore, different strategies are required to

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simulate these molecules, *e.g.*, QE-DFT or quantum embedding, with the potential to scale to the systems of interest from the results of smaller systems. Embedding approaches do help with reducing the dimensionality of a system. However, for the inhibitor this is still prohibitively expensive for meaningful choices of fragmentation. By combining quantum embedding with QE-DFT, however, inhibitor and Michael acceptor fragments can be used to provide data for the QE-DFT training routine.

Results

Herein, we highlight some of the results that we have achieved in the three phases of the Q4Bio challenge program, aiming to be accessible to the non-specialist. More technical discussions will be presented elsewhere. It is widely appreciated that quantum chemistry is an obvious and early application area for quantum computing. More specifically, it is the quantum chemistry of strongly correlated systems where there is the prospect of quantum advantage. In a molecular setting, exemplars include metal centres in proteins [23] and multi-excitons in a light-harvesting complex from purple photosynthetic bacteria [24]. A critical question, thus, is: what is the significance of electron correlation in the system of interest?

Electron correlation

There are several methods for measuring the level of correlation present in a molecule. Here, we discuss one approach, where we compare a so-called multi-reference active space approach, which includes electron correlation, with a mean-field approach (Hartree Fock), that does not include correlation. An active space is defined by selecting a subset of orbitals that are partially occupied or close to the highest occupied and lowest unoccupied molecular orbitals, in the case of the inhibitor molecule they belong to delocalised π-orbitals. In table 1, we compare the inhibitor and cysteine molecule in the equilibrium reactant geometry, and transition state geometry. The Hartree-Fock (HF) weight indicates the contribution of the HF reference determinant to the multiconfigurational wavefunction. A lower weight (*e.g.*, 0.63 for the reactant) suggests significant multireference character, highlighting the importance of strong correlation. The transition state, with a higher HF weight (0.89), is closer but still benefits from a multireference treatment.

Structure	E_{HF}	E_{MP2}	E_{CASSCF}	$E_{\mathrm{SC\text{-}NEVPT2}}$	**HF Weight**
Reactant	-1851.2299	-1853.0996	-1851.4034	-1853.1411	0.63
Transition State	-2730.5141	-2732.8382	-2730.6071	-2732.8519	0.89

Table 1. Absolute energies and the Hartree-Fock (HF) determinant weight from the CASSCF wave function for the inhibitor and S–C$^{\beta}$ bond-forming transition state. Energies shown are from HF, second-order Møller-Plesset perturbation theory (MP2) [25] and strongly contracted *n*-electron valence state perturbation theory (SC-NEVPT2) [26–28] calculations. MP2 and SC-NEVPT2 used the HF and CASSCF wave function as their reference, respectively.

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The active spaces considered here represent the minimal subspaces required to achieve a qualitatively correct description of the bond formation. However, additional electrons within the inhibitor molecule, not currently included in the active space, reside in other conjugated π systems. A comprehensive treatment would ideally incorporate these electrons, but the resulting active spaces might become prohibitively large for classical methods.

For example, modern multireference methods, such as density matrix renormalisation group (DMRG) [29, 30], enable accurate treatment of active spaces of the order of 30 to 60 orbitals, the active space approaches do not recover all dynamical correlation energy, due to the unphysical division of orbital subspace into active and inactive. The subsequent incorporation of dynamic correlation through post-DMRG methods introduces severe computational bottlenecks as high-order reduced density matrices (RDMs) are required, leading to prohibitive scaling for systems where the full-valence π-active space grows rapidly.

It is against this backdrop that we have been motivated to explore the question of can quantum computers help with obtaining more accurate descriptions of inhibitor molecules?

Quantum algorithm development

In Phase I, the objective of the programme was to establish the quantum algorithm to be used to address the Healthcare question, *i.e.*, our drug discovery programme for the treatment of DM1. This required us to evaluate the algorithms available to use, understand the quantum hardware resources requirements and innovate through improvements in, or creation of, algorithms. To this end, we made quantum algorithmic advances within a practical integrated framework, which represents a significant step forward for the potential of quantum simulation of novel molecules in drug discovery.

In the initial stage of the process, classical simulation is used to produce molecular Hamiltonians as input to the quantum workflow. These are, in turn transformed into encoded Pauli Hamiltonians (by using proprietary encodings, fermion-to-qubit mappings, *etc.*) and then used to calculate quantum resource estimates for ground-state computations or to build circuits for emulation or implementation on real hardware (Figure 3). When it comes to quantum circuits, we often use the metrics width and depth, which correspond to the number of qubits and number of gate operations we wish to implement, *i.e.*, we need to target systems that can fit on our device's number of qubits, and we also have to have shallow circuits to minimise error and limit decoherence.

Initially, we evaluated advanced embedding methods for molecules, implemented generalised encoding (GE) as a means to represent our quantum problem on quantum hardware, by tailoring a given molecule to a specific qubit Hamiltonian, and created quantum-enhanced DFAs with polynomial scaling. The GE technique was inspired as a generalisation of the Derby-Klassen [31] compact encoding originally developed and applied to a study of materials [32], and subsequently adapted for our molecular application. In the cases of ethene and acetaldehyde as two prototypical systems, we found circuit depths to be on the order of between 10^3 and 10^5 2-qubit gates, depending on the precise setting, for one layer of the Hamiltonian variational *ansatz* (HVA) [33]. The 2-qubit gates are generally considered to be the largest source of noise within near-term devices and it is critical to minimise their number. Unfortunately, our initial attempts meant the count was still too large

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for simulation on available hardware. So we needed to investigate other strategies to reduce the complexity.

One such approach is to engineer the Hamiltonian of the physical problem through a method such as DMET, which at a high-level can be considered a divide-and-conquer method. In Table 2, we use butadiene in the minimal, STO-3G, basis set to illustrate further the effectiveness of our strategies, considered in the early stages of the project to reduce the resources required using the HVA and the traditional unitary coupled cluster with singles and doubles (UCCSD) *ansatze* [34]. Without employing any fragmentation schemes, the full electronic structure calculation of butadiene with the minimal basis set involves 52 spin orbitals, and thus requires 52 qubits for simulation on a quantum computer. By utilising a suitable fragmentation scheme via DMET [20], the electronic structure problem to be solved is reduced in size, which in turn trims down the required circuit width. The circuit depth can be reduced by another order of magnitude by replacing the UCCSD *ansatz* for the electronic structure problem with HVA, and by also relying on the GE scheme to represent the Hamiltonian in qubit space rather than the Jordan-Wigner mapping [35].

Calculation	**Encoding**	***Ansatz***	**Filtering (%)**	**Qubit Count**	**Circuit Depth**
Full (52 orbitals)	JW	HVA	0	52	2.3×10^{6}
Full (52 orbitals)	JW	HVA	1	52	2.7×10^{4}
DMET (28 orbitals)	JW	UCCSD	0	28	3.7×10^{5}
DMET (28 orbitals)	GE	HVA	1×10^{-9}	28	4.8×10^{4}
DMET (28 orbitals)	GE	HVA	1×10^{-2}	28	2.5×10^{4}

Table 2. Resource requirements for a full active space treatment of butadiene in the minimal, STO-3G, basis using DMET. Encoding schemes included are Jordan-Wigner (JW) and generalised encoding (GE).

As we progressed in the project, it became evident that we needed further resource improvement for our algorithms. To this end, other *ansätze* were studied, and we found the unitary cluster Jastrow (UCJ) *ansatz* [36] to provide significant resource benefits for systems of interest. Table 3 shows some estimates for water, the so-called 'universal solvent' that is

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essential for biological life on Earth, and ethene, which serves as a prototype for strongly-correlated studies when examining its internal rotation. Both of these systems are amenable to exact simulation, which can provide a benchmark for our approaches. Butadiene, acetaldehyde and methyl vinyl ketone are beyond the abilities of exact classical simulation and represent prototypes for our initial studies and a Michael acceptor, respectively. In comparison to the HVA values resources (Table 2), we find an order of magnitude improvement over both HVA and UCCSD. Where exact energy values are available for benchmarking (water and ethene), the number of layers required to approach chemical accuracy (1.5 mHa) is found to be 1 and 4, respectively.

Within the simulation pipeline, our workflow does not end at the quantum simulation as there is a final step which harnesses QE-DFT [19]. This step transfers the accuracy of quantum computation from minimal chemical systems on limited quantum resources to larger, chemically meaningful systems. Using data obtained from the quantum device, DFA training routines are exposed to exchange-correlation energy and densities for systems obtained from a quantum device. Once trained, the resulting DFAs can be embedded in the classical QM/MM to bring advantages in the simulation of activation barriers and binding of the inhibitor within the protein pocket. QE-DFT allows immediate scaling with resource availability and flexibility to extend quantum simulation beyond ground-state energies.

We employ two strategies to fit or train quantum-enhanced DFAs. The first approach is to use a standard algebraic form of the functional with adjustable parameters fitted to the quantum data. This approach allows one to traverse the so-called Jacob's ladder of DFT [37]. We have targeted low rungs for our initial proof of concept studies and also to obtain fast-solving DFAs to embed in the UoN QM/MM pipeline. The second approach we have taken is inspired by Google DeepMind's DM21 approach to neural network functional training, where a composite functional form is weighted by the neural network parameters [38]. It should be noted that we are working with small data sets at present, and are not looking to develop a universal DFA for general applicability. Rather we are developing bespoke DFAs that are well suited to a particular problem or class of systems, *e.g.*, the covalent inhibitors, aiming to improve on known limitations such as the prediction of activation barriers (Figure 7).

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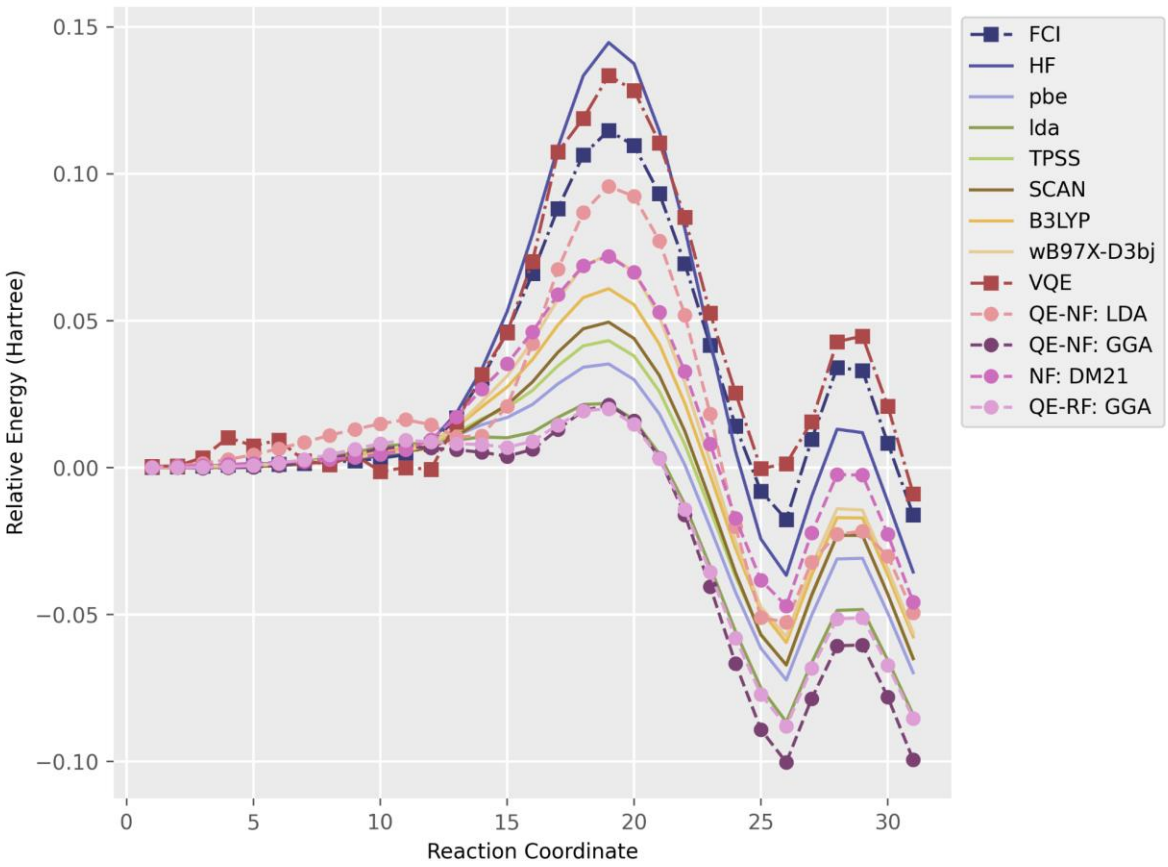

Figure 7: Energy profiles for the concerted HCN + HF → HC(NH)F reaction (with bond formation and internal hydrogen rearrangement) calculated using different methods. The quantum-enhanced DFAs are trained to a subset of data and then used to reproduce the energetics of the nudged elastic band images along the reaction pathway. It can be seen on this proof-of-concept dataset that we are able to generate bespoke DFAs that can provide a better description of the reaction pathway than other leading or commonly used DFAs.

In order to test the performance of new quantum algorithms for simulating covalent inhibitors, benchmark reference data are needed. Thus, we have used high-level quantum chemistry calculations, up to CCSD with perturbative triple excitations [CCSD(T)] [39], to characterise the nucleophilic addition of an archetypal nucleophile, methanethiolate, to various nitrogen-containing Michael acceptors which are representative of the small-molecule covalent inhibitors [40]. Using a hybrid QM/MM approach on conventional HPC, with D3(BJ)-corrected PBE [41] as the QM method (where D3(BJ) is the DFT-D3 dispersion correction [42] with Becke-Johnson damping [43]), we have computed the free energy surfaces, for the binding of several covalent inhibitors to the Cys1039 residue of CDK12.

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Molecule	No. spin orbitals	No. two-qubit gates	No. variational parameters
Water	14	98	107
Ethene	28	304	254
Acetaldehyde	38	434	359
Butadiene	52	616	506
Butadiene (DMET)	28	304	254
Methyl vinyl ketone	62	746	611

Table 3: Resource estimates for the UCJ *ansatz* on some small molecules considered within the Q4Bio challenge program. Within this *ansatz*, we can choose the number of UCJ layers, the Givens rotation rank and graph connectivity. In general, increasing these values will increase the accuracy but at the cost of quantum resources. The results shown are for one layer of the *ansatz*, using a Givens rotation of rank 4 and a graph connectivity that connects nearest-neighbour qubits within each spin sector and the spin up and down qubits for each orbital.

Quantum algorithm emulation

The main goal for Phase II was to prepare our algorithmic framework developed in Phase I for experimental demonstrations of quantum utility in Phase III. The target for the phase was to perform large scale emulations, in excess of 30 qubits. We used these emulations on classical devices for validation of our quantum algorithms and in preparation for efficient and robust deployment on QuEra's neutral-atom hardware. We performed VQE simulations on molecular systems of up to 34 qubits, with an emulation campaign on graphic processing unit (GPU) hardware garnering an unprecedented amount of VQE data. We used the resulting data to train neural network quantum-enhanced DFAs. VQE and QE-DFT energies were compared to full configuration interaction, CCSD and DFT calculations performed on conventional computing, with promising results in reproducing atomisation energies.

The procurement of an Nvidia H200 server, commissioned at the beginning of Phase II, increased our emulation capabilities from 32 qubits on a single A100 GPU, to 33 qubits

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using a single H200 GPU, or 36 qubits using all eight H200 GPUs in parallel. This hardware has been instrumental in emulating our quantum algorithm on more than 30 qubits. In doing so, we have learnt how to deal with the complexities in executing our algorithms at scale, created a significant database of quantum circuits and their results on chemical systems, and in turn used these data to improve QE-DFT. The emulation of our set of molecular systems with different *ansätze*, thresholding, and circuit depth has provided an unprecedented amount of data. At the end of Phase II, this campaign comprised: 149 different molecule/image pairs ("image" refers to a snapshot geometry sampled from a potential energy surface), 447 quantum circuits/resource estimates (molecule/image/threshold combinations), 368 emulations (molecule/image/threshold/*ansatz* combinations that have VQE data), 70 systems emulated (molecule/image combinations that have VQE data) and over 26,000 VQE iterations. The results from the emulation campaign are summarised in Figure 8.

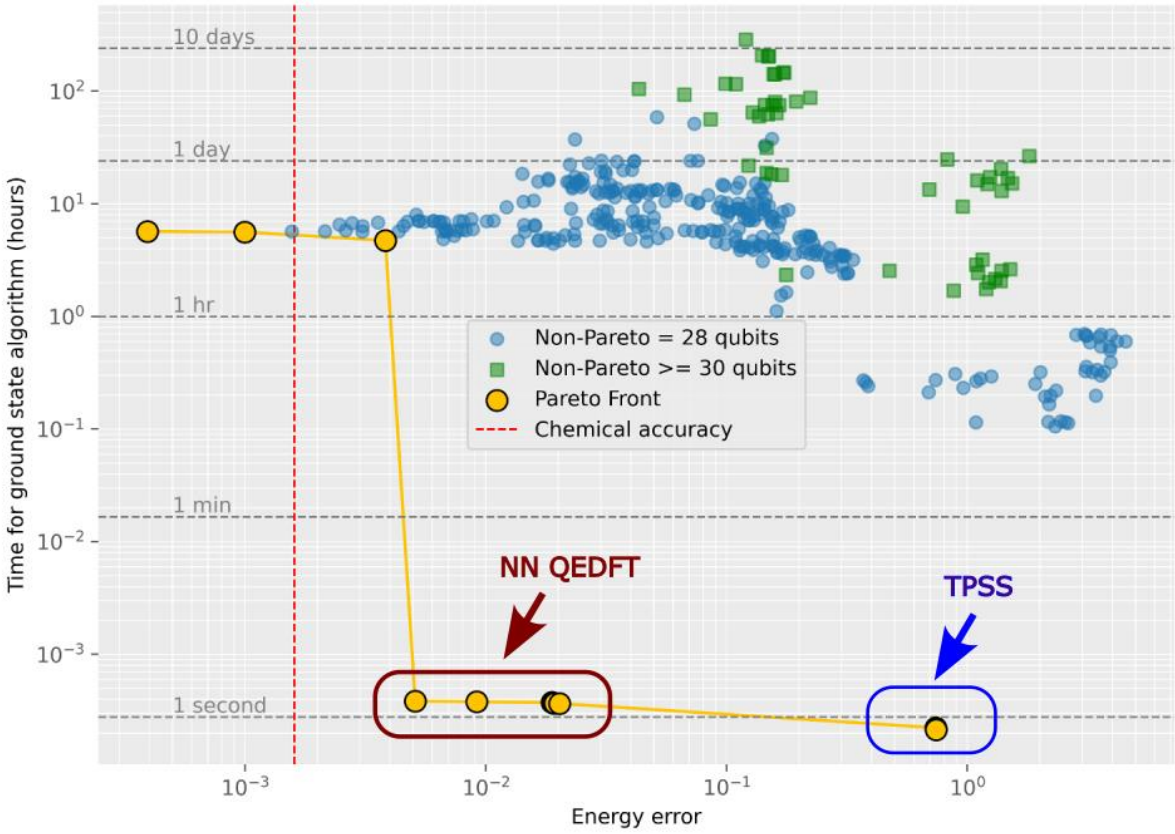

Figure 8: Pareto plot of the energy error versus time for the ground-state solution in the emulations conducted during Phase II. VQE emulations are shown as green and blue markers; the Pareto front is depicted in yellow. Our initial proof-of-concept neural network quantum-enhanced DFA results are highlighted as sitting on the Pareto front optimising for time to solution and DFA. TPSS, a commonly used meta-generalised gradient approximation, is shown for comparison. We can obtain a significant increase in accuracy with only a marginal increase in computational cost.

As we progressed through molecular complexity (number of atoms, number of electrons), *e.g.*, H_2 to LiH to H_2O to C_2H_4 to CH_2O+H_2O, we have faced challenges with the central quantum routines. In overcoming these barriers, we have gained valuable insight into how to

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consistently produce good results from our framework. Running at scale has thus allowed us to improve our state preparation (*e.g.*, through consideration of symmetries), optimise *ansatz* layers and accuracy trade-offs, deploy measurement for both energy and RDM observables, implement automatic differentiation and improve the classical optimiser within the VQE loop (*e.g.*, to handle large numbers of parameters and run within reasonable timescales).

Execution on quantum hardware

QuEra has recently announced a programmable quantum processor operating with up to hundreds of physical qubits called Gemini [44]. It is QuEra's first gate based quantum processor and it features high two-qubit gate fidelities, arbitrary connectivity, as well as fully programmable single-qubit rotations. In Phase III, we will be executing quantum algorithms on Gemini (Figure 9). The detailed specifications of the machine are: up to 260 qubits, all-to-all connectivity, a two-qubit gate infidelity of 0.3–0.5%, a global one-qubit gate infidelity of 0.02–0.05% and a local one-qubit gate infidelity of 0.1–0.3%.

Figure 9. The QuEra Gemini gate-based device being used in our Q4Bio challenge [45].

QuEra's neutral-atom platform has some particular advantages, compared with other alternative modalities, for our purposes. It has a highly flexible architecture that can directly realise long-range entangling gates by moving dozens of atoms simultaneously, avoiding many costly swap gates. The high degree of parallelism available in the gate region enables simultaneous execution of many two-qubit operations. This supports highly time-efficient circuit execution and helps preserve coherence across the full computational sequence.

Before deploying an algorithm on hardware, it is important to be armed with as much information as possible about the behaviour of a device and the likely influence of factors such as noise on an algorithm, from which mitigation choices can be made. To support this, we have been optimising the circuit representations for execution on QuEra's quantum hardware. This includes developing and implementing state-of-the-art ideas, such as (i) warm-starting variational quantum circuits with matrix product states, (ii) novel *ansatz* design, and (iii) exploring circuit compression methods. We have also developed and

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deployed a noise model emulator. Insight garnered from this model will enable us to extract the systems and circuits with the greatest chance of success from our database and marry this with hardware-specific optimisation.

Discussion

Reflections on the Q4Bio *modus operandi*

The initial call for expressions of interest for Q4Bio set out a timetable, which was indicative of the ambitious timelines that Wellcome Leap like to operate within. There was little over a month between the call being issued and the deadline for submission of an Expression of Interest, with feedback provided within 15 days. Full proposals were due 30 days later. One of the advantages of this compressed timeline was that the momentum generated for the initial submission did not dissipate. Thus, even though the feedback that we initially received was not encouraging, the level of commitment required to assemble a new inter-disciplinary team and the enthusiasm of all partners were such that we were confident that we could address the feedback and we were motivated to submit a full proposal. Decisions on full proposals were communicated just one month later. Short timelines throughout the three phases of the Q4Bio challenge program have motivated us to find imaginative approaches to expediting institutional processes that are often less agile than one would like, both for recruiting and retaining researchers on fixed-term contracts and for procuring hardware. Necessity is the mother of invention!

We have benefitted from meetings every couple of months with a Program Director with a high level of relevant specialist expertise and formal reporting on an approximately quarterly basis. In addition, every six months, there have been in-person all-team PI meetings. The all-team PI meeting at the end of Phase I informed the decision on which teams progressed to Phase II. It was held at Ascot, of horse-racing fame, in the United Kingdom and with each team invited to anonymously score the other teams, the venue had a certain resonance! Subsequent all-team PI meetings have fostered some collaborative interactions. For example, several teams, including ours, have been working on a benchmarking effort, which has systematically compiled Hamiltonian data across multiple molecular systems to establish standardised resource estimates for quantum circuit implementations, perform optimisations, and apply approximations up to a predefined desired accuracy.

Outlook

At the start of this project, the impact of quantum computing on Healthcare applications was largely a matter of conjecture. To be useful, advances in computer hardware require concurrent advances in algorithms, as we have seen from classical computing. Within our project, we have brought together a live drug discovery programme for the currently untreatable DM1 (UoN), together with a targeted quantum algorithm development programme (Phasecraft), and cutting-edge neutral-atom quantum hardware (QuEra). The combined effort has demonstrated a computational framework, leveraging a technique known as QE-DFT. A unique driver of this approach is its plug-and-play practicality: computational chemists can use QE-DFT in real drug discovery workflows without quantum computing expertise, bringing the advantages of quantum computing within everyday industrial practice. This is our solution to accelerate towards a future where quantum-enhanced drug discovery becomes a reality.

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Through the phases of the project, we have worked towards the successful demonstration of the NPQ hybrid quantum computing-classical computing framework, culminating in execution on QuEra's Gemini platform, including end-to-end generation and application of a quantum-enhanced DFA within a QM/MM pipeline. The project is closely integrated with a live drug discovery programme for treatments of DM1, where our workflow is extensible to an endeavour needing to simulate and screen thousands of molecules for potency and selectivity. It enhances the specific classical workflow (free energy, umbrella sampling QM/MM molecular dynamics) we seek to improve, and importantly is capable of executing simulations on a timeframe and cost that beats or is commensurate with experimental progress.

Furthermore, our QE-DFT approach is applicable to other biochemical reactions, and, therefore, other ligand-receptor drug discovery and lead optimisation studies. The target of our project, CDK12, is also implicated in a range of other human pathologies. Our approach offers a distinctive pathway for integrating quantum computing into drug-design pipelines by enhancing the core methodology (DFT) that these pipelines already rely upon. While we focused on establishing the fundamental workflow during Wellcome Leap's Q4Bio challenge program, advances in quantum hardware will progressively increase the size of simulable chemical systems and improve the accuracy of quantum outputs, potentially surpassing classical methods. Consequently, our QE-DFT framework can be systematically updated as more capable quantum processors become available. This creates the prospect of using quantum-derived data to train improved DFAs and, ultimately, to support drug design efforts for currently untreatable diseases, exemplified via DM1.

Abbreviations

CASSCF: complete active space self-consistent field

CCSD(T): coupled cluster with single, double, and perturbative triple excitations

DFA: density-functional approximation

DM1: myotonic dystrophy type I

DMET: density matrix embedding theory

DMRG: density matrix renormalization group

GE: generalised encoding

HVA: Hamiltonian variational *ansatz*

JW: Jordan-Wigner

PMF: potential of mean force

RDM: reduced density matrix

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UCCSD: unitary coupled cluster with singles and doubles

UCJ: unitary cluster Jastrow

VQE: variational quantum eigensolver

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